LAB 04_05

Marks = 20

TASK: CIFAR-10 IMAGE CLASSIFICATION USING ANN

Objective

Build a configurable Artificial Neural Network (ANN) for classifying images from the CIFAR-10 dataset and investigate the effect of model architecture, regularization, Batch Normalization, and optimizers.

Task Requirements

1. Load and preprocess CIFAR-10 and visualize representative images with class labels.
2. Build a configurable ANN with adjustable hidden layers, neurons, activation function, dropout, and Batch Normalization.
3. Train the model using at least three optimizers: SGD, Adam, and AdamW.
4. Compare regularization approaches: baseline, L2/weight decay, Dropout, Batch Normalization, and a combined model.
5. Plot and compare training/testing loss and accuracy curves for the selected experiments.
6. Evaluate the best model using Accuracy, Precision, Recall, F1-score, classification report, and a confusion matrix.
7. Use unseen CIFAR-10 test images for prediction and display the actual class, predicted class, and confidence.
8. Display and analyze incorrectly classified images and identify common class confusions.
9. Perform a small hyperparameter study by changing architecture, learning rate, batch size, or other suitable parameters.
10. Select the best configuration and report the final architecture, optimizer, hyperparameters, and test performance.

Expected Deliverable

Submit one Jupyter Notebook (.ipynb) containing the complete implementation, visualizations, experimental comparisons, evaluation results, predictions, and a brief discussion of findings.

Discussion Questions

1. Which optimizer and regularization technique performed best? Did Batch Normalization and Dropout reduce overfitting?
2. Which classes were most frequently misclassied, and why might an ANN be less suitable for image classification than a CNN?